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Get Wired (Again): Trolleybuses and Trolleytrucks

Trolleybuses and trolleytrucks have all the advantages of electric cars – and none of their drawbacks.

July 10, 2009 Written by Kris De Decker Translation fr

Image: A trolleybus in Lyon, France. Image by Florian Fèvre (CC BY-SA 4.0).

A large-scale introduction of electric cars faces many technological hurdles and promises to be time-consuming and expensive. Greening public transportation and cargo traffic, on the other hand, could be done fast with existing technology for a reasonable price - if we opt for the trolleybus and the trolleytruck.

> Cargo traffic and public transport could be electrified in just a few years time.

A trolleybus (or "trackless trolley") can be defined in two ways; as an electric bus that gets its power from overhead cables, or as a tram (or "street car") that drives on rubber tyres. Whichever way you look at it, this combination of bus and tram is the most ecological (motorised) means of transport that exists in the world today.

Just like all other electrically powered vehicles (cars, trains, trams) a trolleybus does not produce exhaust fumes, is more efficient than vehicles with a combustion engine, and can drive on renewable energy. The trolleybus, however, has interesting advantages over other electric vehicles.

Cheap, fast, durable

A trolleybus does not need a battery. In this way, it bypasses the weak point of electric cars. Batteries limit the mileage of electric cars, which means that the vehicles require an elaborate infrastructure for fast-charging or swapping batteries. Batteries also make electric vehicles heavy and thus less energy efficient than when hooked up to an overhead line - a battery makes up at least one third of the weight of an electric car.

A trolleybus also has advantages compared to other means of electric public transport. Contrary to a train or a tram, a trolleybus does not need a rail infrastructure. This not only results in huge cost and time savings, it also saves a large amount of energy for building the infrastructure (see for instance this paper: "Environmental assessment of passenger transportation should include infrastructure and supply chains (http://www.iop.org/EJ/article/1748-9326/4/2/024008/erl9_2_024008.pdf?request-id=be050c6a-1f7b-4d85-83d6-76b91671bf98)", pdf). Installing a trolleybus service is of course more expensive than installing a normal bus line, but that

extra cost can be recovered because of lower fuel and maintenance costs.

Image: A trolleybus in Minsk. Credit: Redline, Wikimedia Commons. (CC BY-SA 3.0)

Furthermore, a trolleybus has better braking power than a tram and it is better at climbing hills, since rubber tyres have more grip than steel wheels on steel rails. Trolleybuses are also compatible with bicycles because cyclists cannot get stuck in the tracks. They are more manoeuvrable than trams - a badly parked car will not stop them, because they can diverge from their track for a couple of metres.

Political advantage

Being public transport, trolleybuses of course have the same advantages as trams; they use much less energy and space per passenger than cars. The trolleybus is not only cheap and ecologically sound, it is also fast to implement. There is no need to break up the road, no need to install a charging infrastructure; just attach overhead lines and off you go. This is a political advantage. The announcement and implementation of a system can happen in the same term of service. Because trolleybuses are cheaper than trams, they can also be used on trajectories where a tram would not find sufficient passengers to be cost-effective.

History and evolution

The first trolleybus got hooked up in 1882; the Elektromote (http://www.obus-ew.de/e4102elektromo.htm) , built by Ernst Werner von Siemens. However, it took almost 20 more years before the first commercial line was installed - in Bielatal, close to Dresden in Germany. During the first half of the twentieth century, and especially since the 1930s, the trolleybus was a success story. Around 1950, there were some 900 trolleybus systems operating worldwide. A large share of these was done away with in the 1960s and 1970s, mostly to the advantage of private cars and diesel buses.

Still, in many cities, the trolleybus never disappeared. Today 359 cities worldwide still operate trolleybus lines, the number of buses is estimated at 40,000. Most trolley services are located in the former Soviet Union and Eastern Bloc countries - probably another reason for their lousy image. The 1,300 kilometre network in Moscow is the largest in the world - it has 1,500 buses and 100 lines.

Image: The first trolleybus, 1882. Image in the public domain.

Image: A trolleybus in England, 1966. Image: Alan Murray-Rust (CC BY-SA 2.0).

Minsk, the capital of Belarus, has the second largest network in the world with 1,050 buses and 68 lines. Saint Petersburg has the fourth largest network in the world with 735 buses spread across 41 lines (following Beijing, China, in third place). Ukraine has trolleybuses in more than 25 cities and it boasts the longest trolley line in the world: 85 kilometres from Yalta to Simferopol (http://www.blacksea-crimea.com/Places/trolleybuses.html) . The three largest networks in the European Union are Athens, Riga and Bucharest. Belgrade, Bratislava, Budapest, Kiev and Sofia are other former Eastern Bloc cities with large trolleybus networks.

Outside Europe

Switzerland has trolleybuses in 13 cities. Dozens of other cities in Europe have smaller networks. Outside Europe there are trolleybus systems in the US (Boston, Cambridge, Philadelphia, Dayton, San Francisco, Seattle), Canada (Vancouver, Edmonton), Central-America (Mexico City, the largest network in the Americas), Latin-America (Argentina, Brazil, Ecuador, Chile) and Asia (China, North-Korea). (sources: 1 (http://de.wikipedia.org/wiki/Oberleitungsbus#Gegenwart) / 2 (http://www.tbus.org.uk/systems.htm) / 3 (http://www.tbus.org.uk/article.htm)).

Obviously, the technology works, because otherwise it would not have been in service for such a long time in so many places. This cannot be said of electric cars, which all but disappeared in the 1920s (https://solar.lowtechmagazine.com/2010/05/the-status-quo-of-electric-cars-better-batteries-same-range/) .

Inferior technology?

Compared to diesel buses, trolleybuses do have a couple of disadvantages. A trolleybus is more manoeuvrable than a tram, but less so than a diesel bus. If the road is being repaired or rebuilt in a street where trolleybuses pass, chances are that the line has to be discontinued temporarily.

A diesel bus can easily be re-routed. Similar to trams, trolleybuses also cannot overtake each other.

The most important drawback of trolley systems is the need for overhead cables. They are generally regarded as ugly and meet protest. Especially at crossroads the cable network can be dense and hard to ignore. Similar to trams, the "tracks" of trolleybuses have points, but the whole mechanism of these hangs in the air. We adore wireless technology and that is probably the reason why trolleybuses are regarded as a ridiculous and inferior technology, a relic from the past.

Image: Trolleybuses in Luzern, Switzerland. Image by Re 460 (CC BY-SA 3.0).

Image: A trolleybus in Milano, Italy.

Hybrid trolleybuses provide an answer to most of these disadvantages. By equipping trolleybuses with a battery or an auxiliary diesel motor, the bus can also cover a part of the route without depending on the overhead cables. Most trolleybuses built since 1990 are equipped with at least a small battery or diesel motor for some limited manoeuvring. This can save the installation of overhead cables, especially at turning points and in sheds, where normally a complicated infrastructure is needed to manoeuvre the buses. It can also help to get round road works.

On some lines (like in Boston and Philadelphia) hybrid trolley services exist. The bus then covers part of the route on electricity delivered by the overhead cables, while another part is covered by means of a (larger) battery or a diesel engine. In this way some drawbacks of batteries and diesel engines are introduced, but these disadvantages are limited when compared to electric cars or diesel buses. Hybrid buses might be a way to spare some parts of a city of overhead lines.

New trolleybus lines

Although some cities have recently decided to stop their (modest) trolley services (Ghent in Belgium, Innsbruck in Austria, Marseille in France and Edmonton in Canada), there are many more cities that have recently expanded or modernised their network, re-introduced trolleybuses, or introduced them for the first time.

In France, the trolley lines in Limoges, Saint-Étienne and Lyon (the largest network in France) have recently been expanded and renewed. One line in Nancy (abolished in 1998) will be restored in 2010. In Athens the full fleet of 350 vehicles has been renewed. In Italy trolleybuses have been re-introduced in Rome in 2005 (only one line) and new systems are coming in Lecce, Avellino en Pescara. The system in Bari will be re-opened. A dozen other Italian cities have never abolished their trolley services and do not have any intention of doing so.

Image: A trolleybus in Lyon.

Image: A trolleybus in Minsk. Image by Romancieslik (CC BY-SA 3.0).

Castellón de la Plana, a city in Spain, re-introduced trolleybuses in 2007, the service was expanded in 2008. In Salzburg (the largest network in Austria with 80 buses and 7 routes) the service was recently expanded. A new system is planned (http://www.tbusleeds.org.uk/) in Leeds in the United Kingdom, which would be the first re-introduction of trolleybuses in the UK in 30 years. Vancouver in Canada renewed its buses in 2007 and 2008, Wellington in New Zealand did the same. Even Ethiopia announced a trolleybus system in 2008.

El Trole

The most spectacular progress is made in South-America. This has everything to do with "El Trole (http://www.publictransit.us/ptlibrary/eltrole.htm)", the trolleybus network in Quito (http://www.tramz.com/ec/q/b1.html) (below), the capital of Ecuador with 1.6 million inhabitants. The already impressive network, built in 1995, was expanded (http://www.tramz.com/ec/q/bm.html) in 2000 and 2008. On a part of the main line (with a length of 19 kilometres) the trolleybuses make use of exclusive lanes, completely separated from other traffic.

During peak hours, there is a bus every 50 to 90 seconds (because of the high frequency, there are no schedules). El Trole transports 262,000 passengers each day. Five other trolleybus lines connect to it, as well as several other bus lines (including Ecovía, a line similar (http://www.nytimes.com/2009/07/10/world/americas/10degrees.html?_r=1&hpw) to El Trole but using diesel buses). The average distance between stops is 400 metres.

Image: A trolleybus stops at a station in Quito, Ecuador. Credit: waldopics, Wikimedia Commons, CC BY 2.0.

The system in Quito is being copied in Mérida (Venezuela), the first part of that line opened in 2007 (picture below). Other cities in Latin America study the possibility of installing a similar infrastructure, and the Quito system was also the inspiration for the proposals in England (http://www.tbus.org.uk/home.htm) and Scotland (http://www.scottishelectrictransit.org.uk/) (second and third picture below).

By choosing the cheaper trolleybus over tram or metro, Quito could develop a much larger network in a shorter time. The capital investment of the 19 kilometre line was less than

60 million dollar - hardly sufficient to build 4 kilometres of tram line (source (http://209.85.135.132/search?q=cache:RfrFv5CxGNMJ:www.sxd.sala.ubc.ca/11_news/tyee_tram_june5_08.pdf+cost+for+building+one+kilometre+tram+line&cd=1&hl=en&ct=clnk&gl=es&client=firefox-a)), or about 1 kilometre of metro line (source (http://www.skyscrapercity.com/showthread.php?p=39000502#post39000502)). Lower investment costs also mean lower ticket fares, and thus more passengers.

Furthermore, the system is well devised (http://www.iges.or.jp/APEIS/RISPO/inventory/db/pdf/0044.pdf) (pdf). There is only one ticket fare, payment happens in the station, not on the bus. Stops are comfortable and built to get fast in and out of the bus, there are very good connections with other lines (sometimes via the same stop), and thanks to the exclusive lanes and (at some crossroads) automatically controlled traffic lights the system is extremely reliable. In Quito, the bus always arrives on time.

Image: A Mérida trolleybus in its exclusive lane. Credit: Joanlink, CC BY 3.0.

Unfortunately, El Trole has become a victim of its own success. The Ecuadorian government now plans to convert (the larger part of) the main line (http://www.publictransit.us/ptlibrary/eltroleupdate1.htm) to a much more expensive light rail line (TRAQ (http://www.innovar-uio.com/web/galerias/traq/TRAQ-book.pdf) , pdf, in Spanish), arguing that the network is saturated. A protest group consisting of citizens and traffic engineers ("Quito para todos (http://www.quitoparatodos.org/) ") opposes the 500 - 750 million dollar plan and demands (http://www.quitoparatodos.org/index.php?option=com_content&view=article&id=32:no-al-tren-ligero-traq&catid=14:documentos&Itemid=24) that the money is used to extend of the trolleyline instead:

"The same investment required to build the 20 to 30 km of light rail would build 250 km of exclusive lanes for trolleybuses including vehicles, stations and terminals. Quito's system of rapid urban mass transport would be complete, providing efficient service, with money left over for construction of bikeways throughout the city, for recovery and integration of public spaces, widening of sidewalks, planting trees and providing urban furniture, building walkways between bus stops and passenger destinations, and other projects to complement the system, in such a way to be able to have a city with an optimal public transport service, placing us in the lead among cities with the best public transport in the world."

Whatever the outcome in Quito will be, the many advantages of a trolleybus line should not lead to the conclusion that light rail systems are evil or unnecessary. When passenger capacity grows, it can make sense to convert the busiest trolleylines to light rail systems. The income of a popular trolleyline might serve to finance the succeeding rail network. Another compromise: rail-guided trolleys (http://en.wikipedia.org/wiki/Bombardier_Guided_Light_Transit) . These vehicles have rubber tyres but are guided by one rail in the middle, which makes it possible to use longer vehicles.

Trolleytrucks

Trolley systems can also be used for the transport of goods. "Trolleytrucks" are a lesser known technology but have an equally long history. Initially, they were as popular as trolleybuses, transporting goods between factories and train stations. Especially the German engineer Max Schiemann put together some remarkable examples (https://solar.lowtechmagazine.com/2009/07/electric-road-trains/) in the beginning of the 20th century.

> There are more elegant options than trolleytrucks, like underground freight networks. Cost, however, is a serious obstacle.

The technology never really took off, though. Trolleytrucks are still sporadically used in Russia and Ukraine (pictures (http://englishrussia.com/?p=736)), and in the mining industry (below, more pictures (http://hutnyak.com/Trolley/trolleyhistory.html#InternationalSalt)). However, in the latter case, the electric engine does not replace the diesel engine, but merely assists it.

Image: A trolleytruck in the mining industry.

Another historical example is the "Valtellina Dam Project (http://hutnyak.com/Trolley/trolleyhistory.html#Valtellina) " in Italy (below). These two lines with a total length of 80 kilometres were built in 1936 and remained in service until 1962. Twenty trolleytrucks transported concrete, sand and other construction materials to build two large dams.

Image: The Valentina trolley system (1938-1962). A total of twenty trolley trucks were used to carry concrete, sand and equipment for the construction of two dams in northern Italy.

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Image: A small trolleytruck in Barcelona, 1956. Image: Dewi Williams.

Today there are no cities that plan a trolleytruck system, but the German city of Dresden does have a Cargo Tram (http://www.railfaneurope.net/pix/de/trams/Dresden/CarGoTram/pix.html) (see

below, it is also being tested in Amsterdam (http://www.citycargo.nl/over.htm)).

Image: A cargo tram.

Trolleytrucks and trolleybuses are also put forward (http://www.tbus.org.uk/trolleylorry.htm) as a solution in the 2008 book "Transport Revolutions: moving people and freight without oil". Authors Richard Gilbert and Anthony Perl propose a plan that would include 500 billion tonne-kilometres of cargo moved by "trolleylorry" in the US by 2025. Trolleytrucks would replace trucks, and complement cargo trains.

High-tech alternatives to trolley systems

There are more elegant options than trolleytrucks, with the same advantages, like the underground freight networks (https://solar.lowtechmagazine.com/2008/02/a-world-without-trucks-underground-freight-networks/) we discussed before. Cost, however, is a serious obstacle. Another alternative for both trolleybuses and trolleytrucks are (wireless) electric buses and trucks (http://www.theoildrum.com/node/5263) , but they too will always be much more expensive, and also less efficient than trolleys - then we are talking about batteries again.

If a bus or truck has a mileage of 100 kilometres, and you have to drive 120 kilometres, you are in trouble. This problem can be solved in two ways. You can put more batteries in your vehicle, but then you increase the cost and the weight and you lower the cargo or passenger space. Or you can set up fast-charging stations or battery swapping stations (https://solar.lowtechmagazine.com/2009/03/who-killed-the-electric-grid-fast-charging-electric-cars/) along the way, but then you increase the costs even more. It gets worse when you start thinking of wirelessly charged buses and trucks. This is a technology that no doubt appeals to more people than trolleybuses do, but it will always be less efficient and more expensive.

All too often we are blind for the costs of high-tech. If we cannot afford a technology, it is of not much use. Low-tech options that have been proven to work can deliver much better results for a bargain. The technology to completely electrify land based transportation has been available for over a hundred years. If we want to, we can do the switchover in just a few years time. Let's start with public transport and cargo traffic, and then let's see what to do with cars - if we still need them.

Trolleycars (https://solar.lowtechmagazine.com/2008/01/why-the-electric-car-has-no-wireless-future/) , even though theoretically possible, are not a practical option.

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